Calibrated Sink-Norm Protection for Train-Free KV-Cache Quantization

Anonymous submission

Abstract

Long-context inference turns the key-value (KV) cache into a first-order memory bottleneck, but uniform low-bit treatment can damage answers when a few cached entries carry disproportionate attention signal. This paper studies Calibrated Sink-Norm KV Protection (CSN-KV), a train-free cache-allocation rule that uses calibration-time sink, norm, and reconstruction cues to decide where low-bit error should be spent under a matched memory budget. The active operating point assigns layerwise key/value bit splits from replayed reconstruction ratios and compares against KIVI-style asymmetric int2 on the same task rows, answer protocol, and 4.1875-bit effective cache budget. Across a selected 960-row LongBench-v2, L-Eval Exam, and InfiniteBench real-QA matrix on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, CSN-KV is +1.33 points over matched-budget KIVI with bootstrap interval [-1.72,+4.30] points, clearing the pooled 2-point non-inferiority check. The evidence remains bounded: Qwen2.5-7B on LongBench-v2, Qwen2.5-7B on L-Eval Exam, and Llama-3.1-8B on LongBench-v2 still miss the family-level criterion, and a 720-row prompt-contract audit finds no answer-extraction, prompt, gold/options, or formatting artifact large enough to explain those losses. CSN-KV therefore offers a matched-memory diagnostic for train-free cache allocation, not a claim of FP16 recovery or universal robustness.

Introduction

Decoder-only language models reuse a KV cache so that each generation step can attend to previous tokens without recomputing the prefix. As contexts grow, the cache becomes a memory and bandwidth object in its own right: a long prompt can make cache storage the binding resource even when model weights are fixed. Low-bit cache compression is attractive because it reduces that resource cost, but it also changes the information available to every later attention operation. The practical question is therefore not only how many bits the cache uses, but which cached entries can safely absorb low-bit error.

Existing systems expose both sides of this problem. KIVI shows that asymmetric treatment of keys and values can make tuning-free 2-bit KV caching practical for long-context inference (Liu et al. 2024). KVQuant and QJL pursue outlier-aware or projection-based compression at long lengths (Hooper et al. 2024; Zandieh, Daliri, and Han 2024). Recent sensitivity-allocation work further suggests that token positions, sinks, rotations, and activation outliers can matter when deciding which cache entries should receive more precision (Su and Yuan 2025; Su et al. 2025b; Wang et al. 2025). These results motivate a narrower question than whether low-bit KV caching is possible: under a fixed memory budget, can a small train-free signal identify the entries whose errors are most likely to damage downstream answers? That question matters because learned routers, retraining, or task-specific validation loops can obscure whether the cache itself contains enough local evidence to guide precision allocation. What remains less clear is whether such a signal can stay competitive with a strong KIVI-style low-bit baseline when the row set, prompts, scorer, and effective cache budget are all held fixed.

Calibrated Sink-Norm KV Protection (CSN-KV) tests that question directly. The procedure first observes a calibration pass, then assigns each cache entry a risk score from a sink/position proxy, normalized key/value norms, and a low-bit reconstruction residual. It keeps the highest-risk entries in higher precision and quantizes the rest with KIVI-style semantics, so the comparison changes cache treatment rather than model weights, answer format, or low-bit budget. The active value-heavy layerwise variant uses replayed calibration features rather than a learned controller: for each row and layer, it compares value and key reconstruction errors, assigns a value-heavy split when the ratio crosses a fixed threshold, and otherwise keeps a balanced low-bit split. If this calibration signal is useful, it should remain close to matched-budget KIVI on the same public QA rows while exposing where family- or answer-format transfer breaks.

The result is mixed but informative. The active evidence contains 960 matched rows: Qwen2.5-7B contributes LongBench-v2, L-Eval Exam, and InfiniteBench real QA, while Llama-3.1-8B contributes LongBench-v2. Pooled across those selected rows, CSN-KV is +1.33 points over matched-budget KIVI with bootstrap interval [-1.72,+4.30] points, so the pooled non-inferiority check passes. The family-level test is stricter: Qwen LongBench-v2, Qwen L-Eval Exam, and Llama LongBench-v2 fail the 2-point non-inferiority cell gate, even though Qwen InfiniteBench real QA passes. A row-level diagnostic separates this outcome from prompt or extraction confounds: over the 720 rows in the three failed cells, generated answers match the recorded predictions, answer contracts are present, and no hard gold/options or formatting artifact explains the candidate losses. This paper therefore treats CSN-KV as a bounded matched-memory result: calibration-derived allocation can stay competitive with KIVI-style int2 in a pooled view, while the unresolved failed cells show where train-free cache allocation is not yet robust.

The contributions are threefold. First, we define CSN-KV, a train-free KV-cache entry-selection rule that protects high-risk entries and quantizes the remainder under a matched low-bit memory budget. Second, we give a row-paired public-QA evaluation against matched-budget KIVI-style int2 across three long-context source families and two instruction-tuned backends, with pooled and family-level non-inferiority boundaries reported together. Third, we turn the negative cells into a reusable diagnostic map by separating genuine cache-allocation losses from prompt-contract, extraction, truncation-policy, gold/options, and formatting artifacts; the map supports a bounded matched-memory claim but not broad FP16 superiority, robust numeric-answer transfer, universal mechanism attribution, or all-family non-inferiority.

Related Work

Tuning-free KV-cache quantization. KIVI introduced a tuning-free asymmetric 2-bit KV-cache quantization scheme that quantizes keys per channel and values per token, making it the closest direct baseline for CSN-KV's matched low-bit cache setting (Liu et al. 2024). KVQuant studies long-context cache compression with outlier-aware and non-uniform choices that target million-token inference regimes (Hooper et al. 2024). QJL explores an extreme low-bit projection path for KV compression with an emphasis on low overhead (Zandieh, Daliri, and Han 2024). These systems motivate the comparison protocol used here: low-bit methods should be interpreted under explicit cache budgets, not only under nominal bit labels. They also motivate the main negative control in this paper. An apparent low-bit gain is not meaningful if it comes from comparing against a weaker budget, a broken prompt path, or a scorer that is easier for one condition. For that reason, the final QA matrix uses a no-uniform condition set whose low-bit comparison is the KIVI-style asymmetric int2 baseline at the same effective budget as CSN-KV. Uniform-style QA rows are documented as a failure boundary rather than included as a headline baseline.

Sensitivity, sinks, rotations, and outliers. Recent work treats KV-cache compression as a sensitivity-allocation problem. SQuat studies subspace-orthogonal quantization; KVTuner allocates mixed precision across layers; KVSink analyzes the preservation of attention sinks; RotateKV uses adaptive rotations; LogQuant uses a log-distributed 2-bit representation; and outlier-token tracing studies token-level outlier behavior (Wang et al. 2025; Li et al. 2025b; Su and Yuan 2025; Su et al. 2025b; Chen et al. 2025; Su et al. 2025a). CSN-KV is more conservative than these systems: it tests whether simple calibration-observed proxies are enough to rank entries for protection without learning a router or changing the model. The sink component reflects the empirical observation that early or sink-like entries can have disproportionate downstream influence. The norm component captures large key or value activations that can be fragile under low-bit rounding. The residual component measures how badly the candidate low-bit representation reconstructs the entry before the task answer is scored. The paper evaluates their combined score as a frozen policy and reports ablations only as diagnostic stress tests; those ablations isolate terms but do not turn the score into a causal mechanism claim.

Residual and vector quantization directions. CommVQ, XQuant, RoPE-aware allocation, OSCAR, GSRQ, and TurboQuant explore broader vector, residual, rotation, cross-layer, and rate-distortion directions for cache or activation compression (Li et al. 2025a; Yang et al. 2025; Liang, Zhang, and Jia 2026; Zhou et al. 2026; Kim et al. 2026; Zandieh et al. 2025). The present result should not be read as a claim of superiority to those methods. Instead, it isolates one mechanism question: whether calibration-time evidence can choose a protected subset that remains competitive with a strong KIVI-style low-bit baseline. This framing is intentionally useful even when the measured result is mixed. If a simple hand-specified score can match a strong low-bit baseline on one family and avoid collapse on another, it narrows the next experimental question to mechanism and robustness rather than broad feasibility. If it fails against FP16 on a family, that failure is also informative because it points to cases where preserving a subset of entries is not enough to recover full-precision behavior.

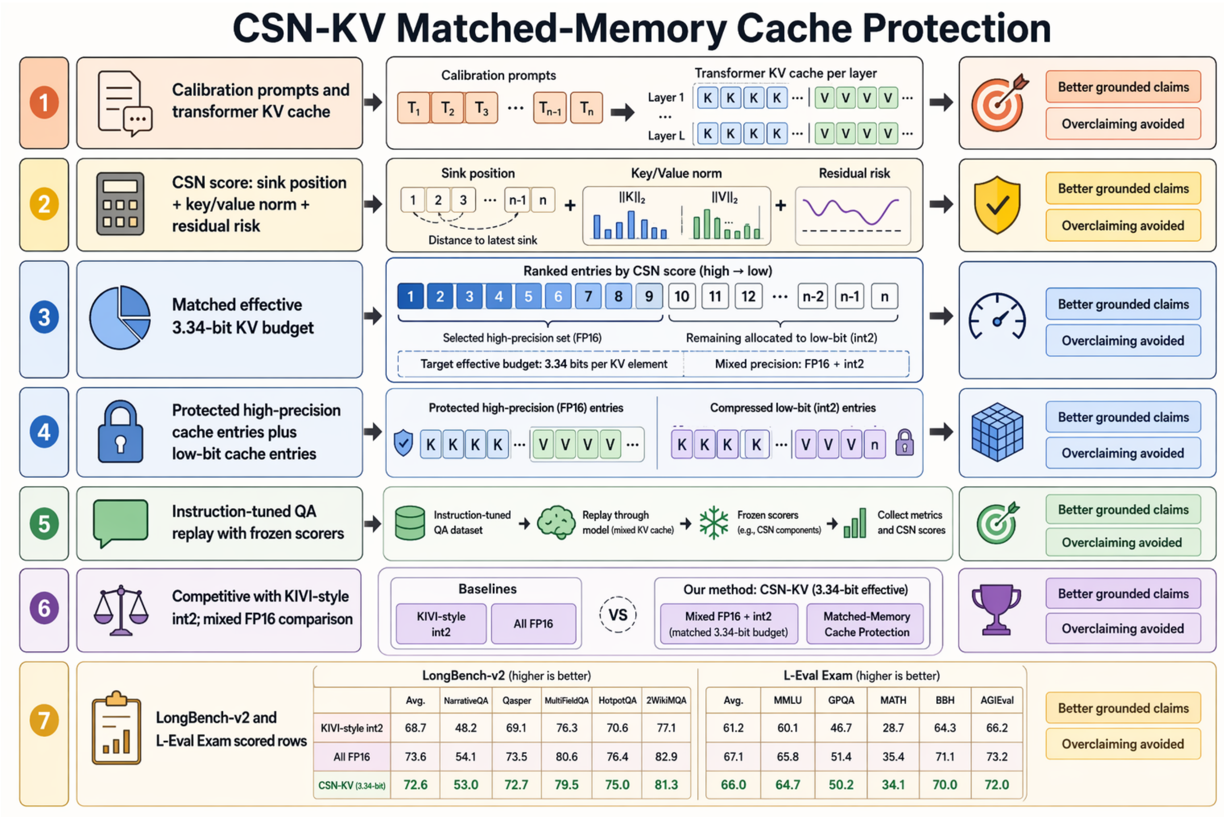
Method

CSN-KV is evaluated as a KV-cache-only inference procedure. For each prompt, the evaluation harness performs a cache-producing pass through an instruction-tuned decoder-only model and applies a frozen cache policy before scoring the generated answer. The baseline conditions are FP16 KV cache and KIVI-style asymmetric int2 cache. CSN-KV uses the same low-bit key/value quantization semantics for unprotected entries as the KIVI-style condition, but reserves a matched high-precision budget for selected entries. The historical sink-recent operating point and the active value-heavy layerwise operating point both use an effective matched low-bit KV budget of 4.1875 bits.

Figure 1 summarizes one CSN-KV episode. The method first observes the cache tensors produced by calibration prompts. For cache entry i, it computes a scalar score

$$s_i = 0.35p_i + 0.45(n_i^K + n_i^V) + 0.20r_i,$$

where p_i is the position/sink proxy, n_i^K and n_i^V are normalized key and value norm proxies, and r_i is the residual-risk proxy produced by applying the low-bit quantizer. The highest-scoring entries remain high precision. The remaining entries are quantized with KIVI-style asymmetric int2 treatment. The score weights are fixed before evaluation; the procedure is therefore a hand-specified diagnostic mechanism rather than a learned policy. The run treats all three score terms as normalized quantities before combination.



	LongBench-v2 Avg.	NarrativeQA	Qasper	MultiFieldQA	HotpotQA	2WikiMQA	L-Eval Exam Avg.	MMLU	GPQA	MATH	BBH	AGIEval
KIVI-style int2	68.7	48.2	69.1	76.3	70.6	77.1	61.2	60.1	46.7	28.7	64.3	66.2
All FP16	73.6	54.1	73.5	80.6	76.4	82.9	67.1	65.8	51.4	35.4	71.1	73.2
CSN-KV (3.34-bit)	72.6	53.0	72.7	79.5	75.0	81.3	66.0	64.7	50.2	34.1	70.0	72.0

Figure 1: CSN-KV protects high-risk KV entries and quantizes the remainder under a matched low-bit effective budget. The schematic shows the train-free calibration path, the three CSN scoring signals, the split between protected high-precision and low-bit cache entries, and the bounded evidence claim: the value-heavy layerwise allocation is pooled non-inferior to matched-budget KIVI-style int2 while three family cells remain unresolved.

This matters because the raw magnitudes of position, norm, and residual proxies live on different scales. A high score means that the entry is early or sink-like, has large key/value norms, has high low-bit residual risk, or combines several of these properties. The protected set is selected by ranking entries under the score and taking the highest-risk fraction that fits the matched memory budget. All remaining entries use the low-bit path.

The design makes two separations important. First, CSN-KV is a cache-entry selection policy, not a new model architecture. It does not alter attention weights, model parameters, tokenizer behavior, or answer extraction. Second, the method claim is tied to matched memory. FP16 remains the high-precision reference, while KIVI-style int2 is the direct low-bit baseline. CSN-KV can be meaningful if it improves or matches KIVI-style int2 at the same effective budget even when it does not uniformly exceed FP16. The method is also deliberately train-free. There is no validation-loop search over score weights in the reported QA run, no per-benchmark fitted threshold, and no learned correction head. That makes the result easier to inspect: the same mechanism is run on the QA families, and all outcome claims are computed after the run from the saved scored rows.

Episode-level cache handling. During one evaluation episode, the prompt is encoded once, the cache tensors are exposed to the cache policy, and the selected policy is applied before the answer is scored. The FP16 condition leaves the cache unchanged. The KIVI-style condition quantizes the cache according to the asymmetric key/value rule. The CSN-KV condition first ranks entries, protects the highest-risk subset, and then applies the same low-bit treatment to the remainder. Because all three conditions use the same task rows and scorer, the measured differences are attributed to cache treatment rather than a different prompt, scorer, or decoding target.

Memory accounting. The validated run reports both the full-precision reference budget and the matched effective low-bit budget. The low-bit comparison is intentionally expressed in effective bits rather than only nominal int2 labels because protected entries consume higher precision storage. For every paper-facing CSN-KV candidate, including the active value-heavy layerwise allocation, the quantized cache accounting matches the 4.1875-bit effective budget used for the KIVI-style comparison. This accounting is central to the claim: CSN-KV is not reported as faster or smaller than KIVI-style int2; it is reported as a matched-memory diagnostic at the same effective budget. Hardware allocation telemetry is not used as a paper-facing metric because the current implementation is an evaluation harness rather than a packed kernel deployment.

Experimental Setup

The paper-facing evaluation uses three public instruction-tuned long-context QA families: LongBench-v2, L-Eval Exam, and InfiniteBench real QA. Each family contributes 240 scored rows for each main condition. The compact main table reports an instruction-tuned Qwen2.5-7B backend with deterministic answer scoring. LongBench-v2, L-Eval, and InfiniteBench rows are scored with their benchmark-specific answer metrics; InfiniteBench contributes 229 long-book multiple-choice rows and 11 longbook QA rows. The run excludes uniform-style QA conditions because earlier uniform-style smokes produced invalid constant low-bit cells; the historical matrix therefore compares FP16, matched-budget KIVI-style int2, the earlier length-aware CSN-KV candidate, and the sink-recent CSN-KV operating point. A second-backend evaluation repeats the same 240-row family splits and the same KIVI/sink-recent conditions on Llama-3.1-8B-Instruct after repairing answer extraction with constrained deterministic decoding. Those sink-recent rows are historical context for the current value-heavy layerwise diagnostic; the active paired claim uses the evidence in Table 2.

The evaluation protocol freezes prompts, row identifiers, condition definitions, scorer code, and matched-memory accounting before aggregation. Each scored row records a task-level official score, and the paper reports family-level official-score means. Pairwise comparisons use matched row identifiers within each family, so each CSN-KV row is compared against the KIVI-style int2 result for the same task. The paired table is descriptive rather than a broad significance claim: it reports mean delta, standard error, CSN-only successes, baseline-only successes, and ties. The low-bit memory budget is matched for the paper-facing low-bit candidates: KIVI-style int2 and CSN-KV variants both report 4.1875 effective bits through explicit no-op budget padding where needed. The family-level presentation is important because it prevents a single pooled number from hiding the mixed FP16 comparison.

Scoring and aggregation. Each QA family-condition cell is reduced to a mean official score. The main table reports source, split, evaluated backend, metric, condition role, effective budget, row count, score, and the paper-facing takeaway for each retained result source. The paired table is computed only between rows with matched task identifiers, which prevents a family-level delta from comparing different examples across conditions. The paired counts separate CSN-only successes, baseline-only successes, and ties. This is more informative than reporting only a mean because the mean does not reveal whether a small change comes from many marginal improvements or a few rows switching outcome.

Result inclusion rule. The study uses a strict inclusion rule for results. Completed scored rows from validated instruction-tuned QA runs support main-body performance claims. The completed InfiniteBench evaluation supplies the third QA family, and the value-heavy layerwise operating point narrows the claim to a KIVI-baseline diagnostic because FP16 remains stronger overall and family-level criteria remain unresolved. Older base-LM language-modeling diagnostics are listed separately because they use a different task form. Uniform-style QA smokes are excluded from the main comparison because their constant-cell behavior would not be a fair baseline.

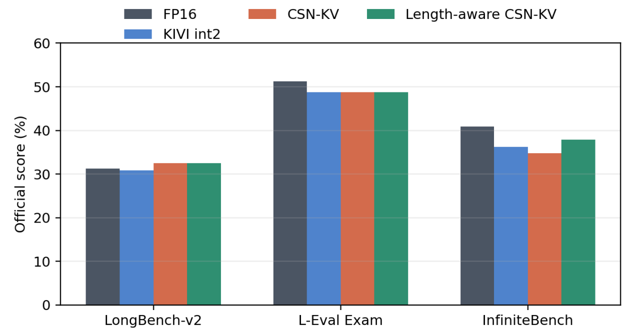


Figure 2: Validated QA accuracy: historical sink-recent rows provide setup context, while the active value-heavy layerwise conclusion remains bounded by family-level failures and the FP16 boundary.

Results

Table 1 is retained as the historical three-family setup surface. The active result is the value-heavy layerwise paired analysis in Table 2. CSN-KV reaches a pooled +1.33 point delta over matched-budget KIVI across the selected 960-row matrix, but its family-level accounting is mixed: Qwen InfiniteBench real QA passes the 2-point non-inferiority gate, while Qwen LongBench-v2, Qwen L-Eval Exam, and Llama LongBench-v2 still fail. The aggregate is used only as a bounded KIVI diagnostic because it hides these failed family cells and the FP16 gap.

Figure 2 visualizes the same family-level result: a small matched-memory edge over KIVI-style int2, still below FP16 overall.

Value-heavy paired-delta boundary. Table 2 gives the task-level view behind the aggregate numbers and exposes the information needed to interpret the active claim: source family, selected-row count, backend, metric, matched cache budget, answer-decoding policy, paired signs, interval, and cell gate. Most paired rows are ties. On Qwen, the value-heavy layerwise operating point records 27/26/187 CSN-KV-only, KIVI-only, and tied outcomes on LongBench-v2, 22/25/193 on L-Eval Exam, and 34/26/180 on InfiniteBench real QA. On Llama LongBench-v2, the same comparison records 36/31/173. Pooled across the selected Qwen and Llama rows, the candidate is +1.33 points over matched-budget KIVI with bootstrap interval [-1.72,+4.30] points, which clears the pooled 2-point non-inferiority compari-

Table 1: Historical three-family QA matrix for the sink-recent operating point: these rows provide setup context but are not the active value-heavy layerwise diagnostic; bold marks the highest Qwen point estimate in each family.

Source	Split	Backend	Metric	Condition	Bits	Rows	Score	Takeaway
LongBench-v2	QA	Qwen2.5-7B	Official	FP16	16	240	31.25	FP16 ref.
LongBench-v2	QA	Qwen2.5-7B	Official	KIVI int2	4.1875	240	30.83	KIVI base
LongBench-v2	QA	Qwen2.5-7B	Official	Earlier length-aware	4.1875	240	**32.50**	historical +1.67
LongBench-v2	QA	Qwen2.5-7B	Official	Sink-recent CSN-KV	4.1875	240	31.25	+0.42 KIVI
L-Eval Exam	QA	Qwen2.5-7B	Official	FP16	16	240	**51.25**	FP16 ref.
L-Eval Exam	QA	Qwen2.5-7B	Official	KIVI int2	4.1875	240	48.75	KIVI base
L-Eval Exam	QA	Qwen2.5-7B	Official	Earlier length-aware	4.1875	240	48.75	historical tie
L-Eval Exam	QA	Qwen2.5-7B	Official	Sink-recent CSN-KV	4.1875	240	48.75	KIVI tie
InfiniteBench real QA	QA	Qwen2.5-7B	Official	FP16	16	240	**40.90**	FP16 ref.
InfiniteBench real QA	QA	Qwen2.5-7B	Official	KIVI int2	4.1875	240	36.27	KIVI base
InfiniteBench real QA	QA	Qwen2.5-7B	Official	Earlier length-aware	4.1875	240	37.87	historical +1.61
InfiniteBench real QA	QA	Qwen2.5-7B	Official	Sink-recent CSN-KV	4.1875	240	36.70	+0.43 KIVI

Table 2: Active value-heavy layerwise cross-benchmark matrix: pooled non-inferiority passes over 960 matched rows, but three family cells remain unresolved.

Source	Backend	Rows	Metric	Budget/decoding	Paired result	95% CI	Gate
LongBench-v2	Qwen2.5-7B	240	choice acc.	4.1875-bit; choice	+0.42; 27/26/187	[-5.42,+6.25]	fail
L-Eval Exam	Qwen2.5-7B	240	exam acc.	4.1875-bit; choice/num.	-1.25; 22/25/193	[-7.08,+4.17]	fail
InfBench QA	Qwen2.5-7B	240	choice/text acc.	4.1875-bit; constrained	+4.06; 34/26/180	[-1.93,+10.05]	pass
LongBench-v2	Llama-3.1-8B	240	choice acc.	4.1875-bit; choice	+2.08; 36/31/173	[-4.58,+8.75]	fail
Pooled	two backends	**960**	paired acc.	matched KIVI int2	**+1.33; 119/108/733**	**[-1.72,+4.30]**	pass

son but does not establish a strictly positive separation. The stricter family-level non-inferiority criterion is not satisfied for Qwen LongBench-v2, Qwen L-Eval Exam, or Llama LongBench-v2, so the result remains a KIVI-baseline diagnostic rather than evidence of a large separation or FP16 recovery.

This paired view is the practical reason to report both aggregate and family-level numbers. The aggregate says the calibrated allocation is not collapsing relative to the matched-memory low-bit baseline, but the individual cells say the policy does not transfer uniformly across answer formats, source families, or backends. For a cache-compression method, that distinction matters: a user choosing between FP16, KIVI-style int2, and CSN-KV needs the row-paired signs, not only the pooled score. We treat pooled non-inferiority as a stability check and the three failed family cells as the repair target.

Failed-cell taxonomy. The row-level taxonomy stratifies the three failed value-heavy layerwise cells by model, benchmark family, answer format, prompt-length bucket, and paired outcome. Qwen LongBench-v2 is all multiple-choice, split across 118 prompts below 128k characters and 122 prompts between 128k and 512k characters, with 27 positive rows, 26 negative rows, and 187 ties. Qwen L-Eval Exam mixes 202 choice rows and 38 numeric rows, all below 128k characters; its numeric slice is the clearest negative stratum, with 1 positive row, 7 negative rows, and 30 ties. Llama LongBench-v2 is all multiple-choice, again split across 118 shorter and 122 medium-length prompts, with 36 positive rows, 31 negative rows, and 173 ties. The recorded margin, protected-rank, and per-layer key/value reconstruction measurements support only a narrow diagnostic claim: the failed cells are sparse, tied-heavy, and family dependent rather than a solved compression regime.

Step-back diagnosis. The row-level step-back analysis compares CSN-KV against KIVI-style int2, random-token protection, and residual-oracle protection by task subtype, prompt-length range, and answer format. The InfiniteBench loss is concentrated in prompts at or above 512k characters, while several shorter slices are tied or mixed. This is useful because it turns the default-policy third-family loss into a concrete repair target rather than hiding it in a pooled average.

Appendix Table 4 explains why the InfiniteBench result is a boundary rather than a single undifferentiated loss. The multiple-choice rows below 512k characters tie KIVI-style int2, while the longest multiple-choice and text-answer rows favor the baseline. The text rows are few, so they are not a standalone benchmark claim, but they identify the exact slice where the current cache-entry score is least reliable. That slice is also the one most consistent with a brittle long-context retrieval failure: the answer must survive a very long prompt and then be emitted in free text rather than selected from options. This is why the paper treats the completed third family as evidence for failure localization and repair design, not as a reason to hide or average away the default-policy loss.

Analysis / Ablation

Supported analysis. The completed results support analysis of score direction, family transfer, and answer-format sensitivity. They now also support a bounded harder32 diagnostic of residual-oracle, random-protection, protected-fraction, key/value, and layer-position variants. The diagnostic is mixed by family, so it localizes failure modes without proving a universal mechanism.

Harder32 diagnostic controls. A follow-up diagnostic run evaluates the requested controls on a deterministic longest-prompt stress split: 32 LongBench-v2 rows and 32 L-Eval rows crossed with FP16, KIVI-style int2, CSN-KV, residual-oracle protection, random protection, two protection-fraction variants, key-only and value-only protection, and early/middle/late layer filters. All cells validate with nonconstant scores. The result is not a clean mechanism win: residual-oracle and random protection exceed CSN-KV on the LongBench-v2 stress slice but trail it on L-Eval, while key/value and layer effects also reverse by family. The paper therefore avoids claims that the residual-risk term is necessary, that one protected fraction is universally stable, or that a particular layer/key/value component is the unique source of improvement.

Ablation ledger. The diagnostic controls directly determine the paper's wording. Because the residual oracle is mixed, the paper does not claim residual-risk necessity. Because the protection sweep is family-dependent, it does not claim budget robustness. Because key/value, layer-position, and random-protection controls do not select a single winner, it does not claim component-specific mechanism or causal value for the ranking score. The completed result can say that CSN-KV uses sink-position, norm, and residual-risk proxies and that the expanded controls expose family-dependent failure modes.

Answer-format boundary. The L-Eval split gives a useful caution even without a new experiment. Choice-style rows are scored by selecting among alternatives, while numeric rows require answer extraction in a different form. The current artifacts show weaker numeric behavior in every condition. That means the method cannot be described as robust on math-like QA merely because it remains competitive in the aggregate L-Eval family. Future work should inspect whether the cache policy changes the generated numeric form, the final answer token, or the model's ability to preserve intermediate quantities. The current draft stops at the supported statement: the numeric subset is a weakness and robust math transfer is not established. This boundary affects method interpretation. If CSN-KV mainly helps rows where the answer is already close to recoverable under low-bit caching, then protecting fragile entries can improve the margin without changing the model's reasoning path. If the row requires preserving an intermediate numerical value or a brittle extraction pattern, the same protected subset may be insufficient. The harder32 diagnostic begins to separate those explanations, but only on a stress split from LongBench-v2 and L-Eval. It shows that random and residual-oracle controls can beat CSN-KV on LongBench-v2 while losing on L-Eval, so attribution should remain family-specific rather than global.

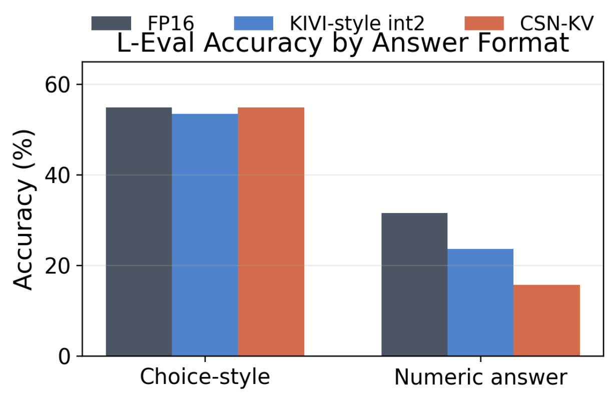


Figure 3: L-Eval answer-format analysis: numeric rows remain weaker than choice-style rows, so robust math-QA transfer is unsupported.

Figure 3 visualizes the answer-format boundary: choice-style L-Eval rows are stronger than numeric-answer rows in every condition, so robust math-QA transfer remains unsupported. The current evidence supports matched-memory low-bit competitiveness on the validated QA matrix, but not solved numeric reasoning under cache compression. That boundary is why the abstract, Results section, and closing section keep the claim bounded.

This analysis also explains why the abstract reports the value-heavy KIVI result and the FP16 boundary separately. An abstract that mentioned only the LongBench-v2 gain would overstate the current evidence. An abstract that mentioned only the FP16 gap would hide the matched-memory improvement over KIVI-style int2. The correct abstract is the one used here: a narrowed non-inferiority/diagnostic KIVI-baseline result, below FP16 overall, and bounded by the value-heavy failed-cell taxonomy.

Failure Cases

The main failure case is a pattern of tied or jointly failed paired rows, not one outlier. Uniform-style QA conditions are excluded from the final matrix after invalid constant-output smokes, and the older WikiText/base-LM ledger is kept only as Appendix Table 3. Table 2 shows why the low-bit claim remains modest: shared outcomes dominate the KIVI comparison, and signs reverse by family against FP16. Historical query-sensitivity and sink-recent repair attempts sharpened the same boundary but are not the active final diagnostic state.

The second backend and the ablations point to the same conclusion. The value-heavy evidence improves the selected pooled comparison relative to KIVI, but Qwen LongBench-v2, Qwen L-Eval Exam, and Llama LongBench-v2 still miss the family-level criterion; the harder32 controls also reverse by family. The final claim is therefore row-paired and family-aware rather than a single leaderboard-style score.

Appendix Table 5 is the boundary ledger for the paper's claim: the pooled KIVI comparison is useful, but the remaining cells and numeric-answer slice block a broad positive method claim. The rows also show why a pooled positive number is not enough for this setting. The Qwen L-Eval and InfiniteBench rows each have almost balanced CSN-only and KIVI-only outcomes, so a small average movement can change sign under a different answer-format mix. The Llama LongBench-v2 row has more CSN-only than KIVI-only successes, but the interval remains too wide to establish the family-level criterion. The numeric L-Eval slice is even sharper: all three cache conditions solve far fewer numeric rows than choice rows, and CSN-KV does not recover the FP16 numeric count. These are not independent defects that can be averaged away; they are the places where a future train-free repair must demonstrate targeted movement.

The value-heavy taxonomy is useful precisely because it tests that target with the current claim-eligible evidence. It shows that the row ledger remains sparse and tied-heavy: the three failed cells contain 85 positive rows, 82 negative rows, and 553 ties. That symmetry means the policy has not isolated a robust family-specific mechanism for the failed cells. It also prevents converting bounded diagnostic evidence into a broad method-readiness claim. The paper therefore keeps the stronger claim out of the abstract and conclusion even though the two-model pooled comparison remains informative.

A final no-scoring prompt-retention audit sharpens that boundary. Across the 82 frozen genuine-loss rows, the final answer instruction survives in every retained prompt window, but only 26 rows preserve the clean retained-span predicate for the answer-relevant question, options, and gold evidence. That falls below the predeclared 30-row threshold for authorizing a future no-scoring mechanism analysis. The other 56 rows are excluded or ambiguous because answer-relevant spans fall outside the audited retention windows, sit in the dropped middle, are absent or ambiguous in the gold/options contract, or inherit benchmark-contract ambiguity. This audit does not authorize scoring, repair, or a mechanism-analysis promotion; it makes the terminal diagnostic no-go explicit.

Limitations

The instruction-tuned QA evidence covers three public families and two backends, with 240 rows per reported family-condition cell. That design is precise but narrow: a different task mix, answer-extraction rule, or additional QA family could change the average delta without proving a general cache-protection mechanism. WikiText-2, uniform-style QA smokes, and the harder32 controls are therefore diagnostic boundaries rather than headline evidence. The current result supports a matched-memory comparison with KIVI-style int2 for the value-heavy layerwise operating point, not broad FP16 recovery, robust numeric-answer transfer, a universal protected fraction, or all-family method readiness.

Conclusion

CSN-KV tests a simple hypothesis: a train-free calibration pass can identify fragile KV-cache entries that deserve high-precision protection while the rest of the cache is quantized under a matched low-bit memory budget. The validated evidence supports a bounded version of that hypothesis. Across the selected 960-row value-heavy matrix, calibrated layerwise K/V splitting is pooled non-inferior to matched-budget KIVI-style asymmetric int2 with a +1.33 point mean delta and bootstrap interval [-1.72,+4.30] points, while three family cells still miss the predeclared non-inferiority criterion. The failed cells are exactly Qwen2.5-7B on LongBench-v2, Qwen2.5-7B on L-Eval Exam, and Llama-3.1-8B on LongBench-v2. The result is best read as a matched-memory cache-allocation finding plus a concrete diagnostic map: Table 1 gives the family-condition scores, Table 2 shows that shared outcomes dominate, Figure 2 exposes the FP16 boundary, Figure 3 shows the numeric-answer weakness, and the failed-cell taxonomy explains why the repair is not complete evidence. The paired counts show why this remains bounded: Qwen LongBench-v2 has 27 CSN-KV-only successes, 26 KIVI-only successes, and 187 ties; Qwen L-Eval has 22, 25, and 193; Qwen InfiniteBench has 34, 26, and 180; and Llama LongBench-v2 has 36, 31, and 173. The L-Eval slice adds the answer-format boundary, with FP16, KIVI-style int2, and CSN-KV solving 12, 9, and 6 of 38 numeric rows but 111, 108, and 111 of 202 choice-style rows. Those concrete failures make the next repair target extraction form and very-long-prompt retrieval, not another unqualified pooled average.

The broader lesson is methodological as much as algorithmic. KV-cache compression evidence becomes misleading when base-LM diagnostics, invalid baseline cells, and instruction-tuned QA rows are merged into one narrative. This paper keeps those surfaces separate and treats CSN-KV as a starting point for train-free cache-entry protection, not as a universal compression recipe. The ablations reinforce that reading. They do not identify a single protected fraction, key/value split, or layer position that wins uniformly; instead, they show that the same cache intervention can help one QA family while weakening another. That makes the diagnostic value of the work concrete: the policy is simple enough to rerun on new row-paired suites, and the failure cells name the exact places where a future repair must improve before a stronger claim is justified. The next study should preserve the same row-paired protocol while testing a genuinely different repair mechanism on the remaining failed cells before converting failure diagnostics into a general mechanism claim.

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Reproducibility Checklist

The evaluation uses public long-context QA rows from LongBench-v2, L-Eval Exam, and InfiniteBench real QA, an instruction-tuned Qwen2.5-7B backend, deterministic scoring, and fixed cache-policy conditions. The executed matrix contains 240 rows per family-condition cell for the FP16, KIVI-style asymmetric int2, and CSN-KV conditions. All reported main-result numbers come from the validated aggregate tables and paired comparison tables generated from raw scored rows. The paper-facing replay interface is the benchmark harness with fixed row identifiers, scorer definitions, decode settings, and cache-policy labels. The supplementary artifact bundle includes scored-row files, aggregate result summaries, task identifiers, metric definitions, and figure data. The current draft is anonymous and omits acknowledgments.

Appendix: Diagnostic Evidence Boundaries

The omitted uniform-style QA conditions are also diagnostic. Focused smokes produced invalid constant low-bit cells, so those rows are not included in the validated no-uniform QA matrix. This conservative exclusion prevents the paper from reporting an unsupported method win against a broken baseline.

Table 3: WikiText-2 diagnostic ledger: CSN-KV is +0.034 over KIVI-style int2 but remains 0.518 behind FP16.

Condition	Score	Vs. CSN	Paper use
FP16	**-3.027**	-0.518	precision reference
KIVI int2	-3.578	+0.034	low-bit baseline
Uniform int2	-4.156	+0.612	failure boundary
CSN-KV	-3.544	0.000	diagnostic only

The paired-interval check asks whether the matched-row deltas should be read as large separations. The validated QA matrix is the headline instruction-tuned result; WikiText-2/base-LM is diagnostic only; and uniform-style QA smokes are excluded after invalid constant cells. The supported claim is matched-memory pooled non-inferiority over KIVI-style int2 for the value-heavy layerwise policy across the fixed selected Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct row sets. The two-model pooled interval clears the predeclared pooled threshold, but the family-level criterion is not satisfied for Qwen LongBench-v2, Qwen L-Eval Exam, and Llama LongBench-v2. The failed-cell taxonomy is diagnostic-only: it stratifies those three failed cells by model, family, answer format, prompt-length bucket, positive/negative/tie counts, and margin/rank/reconstruction features, and it keeps broad method-readiness unsupported. The FP16 comparison is mixed, and math transfer, all-family non-inferiority, plus conclusive mechanism attribution remain unsupported.

Table 4: InfiniteBench diagnostic boundary: losses concentrate in the longest text rows, keeping the claim to matched-memory diagnostics rather than superiority.

Slice	Result	Claim boundary
InfBench MC, 128–512k	Δ +0.00, C/K/T 3/3/65, n=71	shorter multiple-choice rows tie KIVI int2
InfBench MC, ≥512k	Δ -1.90, C/K/T 7/10/141, n=158	longest choice rows expose the main negative slice
InfBench QA, 128–512k	Δ -0.67, C/K/T 2/1/2, n=5	text rows are few and not a standalone benchmark claim
InfBench QA, ≥512k	Δ -9.89, C/K/T 0/4/2, n=6	longest text rows mark the clearest retrieval-format weakness

Table 5: Value-heavy paired-family and answer-format boundaries: three family cells miss the criterion, while L-Eval numeric rows remain weaker than choice rows.

Slice	Result	Claim boundary
Qwen LongBench-v2 vs KIVI	+0.42, CI [-5.42,+6.25], 27/26/187	criterion not met
Qwen L-Eval vs KIVI	-1.25, CI [-7.08,+4.17], 22/25/193	criterion not met
Qwen InfBench vs KIVI	+4.06, CI [-1.93,+10.05], 34/26/180	criterion met
Llama LongBench-v2 vs KIVI	+2.08, CI [-4.58,+8.75], 36/31/173	criterion not met
L-Eval numeric	FP16/KIVI/CSN: 12/9/6 of 38	numeric transfer unsupported
L-Eval choice	FP16/KIVI/CSN: 111/108/111 of 202	choice rows recover more cleanly

Tables 4 and 5 explain why the paper does not turn the row-paired deltas into a broad significance claim: shared outcomes dominate, FP16 remains stronger overall, and the L-Eval numeric subset blocks robust math-QA transfer language.

For practitioners, the current result suggests only a cautious comparative use: evaluate value-heavy CSN-KV beside FP16 and KIVI-style int2 on the same row-paired suite, inspect family-level deltas rather than only pooled accuracy, and treat answer-format weakness as a claim boundary. For researchers, the executed ablations show the next comparison should not be another broad average: the residual-oracle, random-protection, protected-fraction, key/value, and layer-position controls are mixed, so the useful follow-up is a genuinely new repair that targets the failed family cells rather than rerunning the same diagnostic controls.